

tschich Perfect proportions.

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tschich is a simple tool for achieving
perfect proportions on the page —
no matter the paper's dimensions.

This manual was set from
the  Garamontio typeface
on 110 × 165 mm paper
using tschich for proportions.

Introduction

JAN TSCHICHOLD in his essay »Willkürfreie Maßverhältnisse der Buchseite und des Satzspiegels« gives an insight into multiple typographical canons, (that is modes of constructing and placing the type area on the page), that preserve perfect proportions especially for the layout of books, where two opposing pages are always viewed at once.

This small package aims to reproduce these canons to provide a simple solution for creating not only sensible, but also elegant typographical results. To achieve this, tschich comes with the following functions:

#tschich-help	Displays package information and the page's dimensions.
#tschich(a, b)	Returns the margins for nine part division.
#tschich-var(a, b, p)	Returns the margins for division with custom proportions.
#tschich-old(a, b, p)	Returns the margins for the medieval ideal canon.
#tschich-mid(f)	Returns the margins for any type area centered horizontally.
#tschich-dim(f)	Returns the dimensions of any type area.

Using *tschich*

SETTING UP the margins for placing the type area on the page is all about proportion. It must always take the page's width (a) and height (b) into consideration. These measurements, (if not already known), can be displayed by using `#tschich-help` which produces the following:

`< tschich v.o.i.o >` — To set up the margins use:

`#set page(margin: tschich(a, b))`

$a = 110 \text{ mm}$ $b = 165 \text{ mm}$

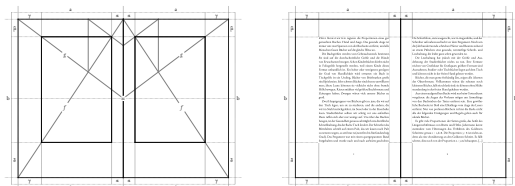
For advanced usage see the  manual.

Given the page's dimensions, the margins can be computed by using `#tschich(a, b)` which returns the values as a dictionary that can be applied as an argument in the set-rule for pages.

This standard function reproduces a nine part division following van de Graaf's adaption of Villard's canon — that is: the inner margin will be one ninth and the outer margin will be two ninths of the page's width. The same goes for the top and bottom margin in relation to the page's height.

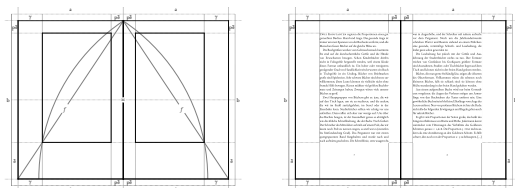
Villard's canon can also be reproduced with a custom proportion (p) by using `#tschich-var(a, b, p)`. For the standard nine part division p would be $\frac{1}{9}$, for a six part division it would

be $\frac{1}{6}$, etc. This proportion must however always be greater than $\frac{1}{3}$, else no type area can be constructed.



Nine part division using Villard's canon.

Furthermore, tschich provides an adaption of the old medieval ideal canon that Tschichold discovered. It can be utilized by using #tschich-old(a, b, p).



Medieval ideal canon with a proportion of one third.

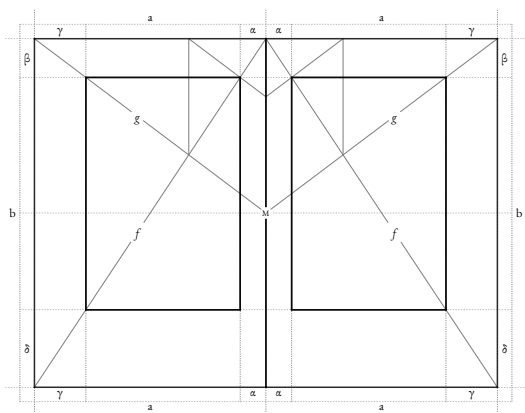
In this case the proportion provided is that of both the inner and top margin in relation to the bottom margin. If these should be a third of the latter, p would be $\frac{1}{3}$, if a fourth, it would be $\frac{1}{4}$, etc.

For all the functions the dimensions of the resulting type area can be computed with `#tschich-dim(f)` where `f` is a call of the function used to construct the margins. This will return a dictionary containing the width and height of the type area. The dimensions of this manual's type area are therefore given by `#tschich-dim(tschich(110mm, 165mm))` for example.

For projects that are to be viewed one page at a time the type area generated by each of the functions can be horizontally centered on the page using `#tschich-mid(f)` which does also return a dictionary of margin values and can therefore be directly applied as an argument in the set-rule for pages.

Geometric Construction

GIVEN THE width (a) and height (b) of the page, construction of the type area using Villard's canon with proportion p begins by taking the midpoint of the page's inner edge (M) to be the origin of a coordinate system with the x -axis pointed towards the outer and the y -axis pointed towards the upper edge of the page.



First the page is split (from its upper-inner to its lower-outer corner) by a diagonal (f) that meets the x -axis in the center of

the page. It can then also be described by the function:

$$f(x) = \left(\frac{-b}{a}\right) \cdot x + \frac{b}{2}$$

Then the page's upper half is also split (from M to its upper-outer corner) with a diagonal g , described by the function:

$$g(x) = \left(\frac{b}{2a}\right) \cdot x$$

The inner (α) and the top margin (β) are to each have proportion p in relation to the page's width and height. Their values are therefore given by:

$$\frac{\alpha}{a} = p \Leftrightarrow \alpha = pa, \quad \frac{\beta}{b} = p \Leftrightarrow \beta = pb$$

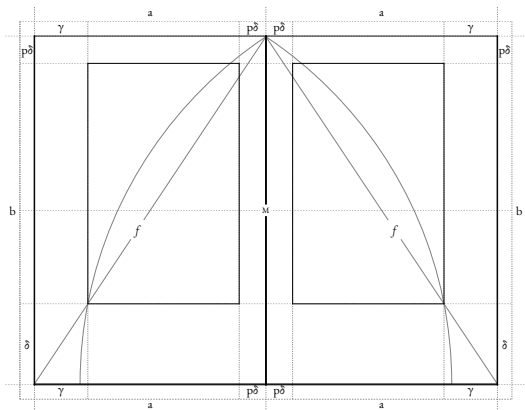
The outer margin (γ) is then the difference of the page's width and the x -coordinate where g meets the top margin, such that:

$$g(a - \gamma) = \frac{b}{2} - \beta \Leftrightarrow \gamma = 2pa$$

At last the bottom margin (δ) is the sum of half the page's height and the y -coordinate where f (which is negative at this point) meets the outer margin, such that:

$$\delta = \frac{b}{2} + f(a - \gamma) = 2pb$$

For constructing the type area using the medieval ideal canon, a circle centered on the lower-outer corner of the opposite page is drawn through the inner-upper corner, such that its radius equals the diagonal of the page.



The x -coordinate (x_c) at which it meets f is therefore given by equating the distance of the intersection and the lower-outer corner of the opposite page to its radius:

$$a^2 + b^2 = (a + x_c)^2 + \left(\frac{b}{2} + f(x_c)\right)^2 \Leftrightarrow x_c = 2a - \frac{4a^3}{a^2 + b^2}$$

The outer margin (γ) is then the difference of the page's width and x_c , such that:

$$\gamma = a - x_c = \frac{4a^3}{a^2 + b^2} - a$$

The bottom margin (δ) is the sum of half the page's height and the y -coordinate of f at x_c (which is negative at this point), such that:

$$\delta = \frac{b}{2} + f(x_c) = \left(2 - \frac{4a^2}{a^2 + b^2}\right)b$$

At last both the inner and top margin are set to each have proportion p in relation to the bottom margin, such that:

$$\alpha = \beta = p\delta$$